

Berklee College of Music

Surround Vocal Spectrum Visualizer

**A Dedicated Plugin Designed to Support the Mixing Workflow for Vocal
Ensemble Music**

Submitted in Partial Fulfillment of the Degree of
Master of Music in Music Production, Technology, and Innovation

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Abstract

This project develops a visual tool in C++ using the *JUCE framework* that represents surround space to support the mixing and mastering process of vocal and choral-oriented music, particularly within a cappella contexts. Variations in timbre, tone, and intonation naturally occur among different vocal groups—such as treble, lower-voiced, and mixed ensembles—due to their unique physiological and acoustic characteristics. As panning and stereo direction are commonly employed in a cappella mixing, it can be challenging for producers to accurately perceive the spatial distribution and balance of individual voice parts within a dense mix.

The software introduces two primary functions: first, to visualize vocal parts in surround space, enabling producers to make more informed mixing decisions and identify locational placement of different mix subgroups; and second, to perform a timbral frequency analysis implementing equalization monitoring to assist in achieving consistent spectral balance across varying vocal timbres. By providing an objective visual framework for both spatial and spectral analysis, the tool enhances clarity, blend, and cohesion in ensemble vocal production while bridging the gap between technical precision and artistic expression.

Keywords: music production, a cappella, vocal mixing, surround sound visualization, spatial audio, multichannel monitoring, C++/ JUCE

This paper was written with the assistance from generative artificial intelligence

Introduction

Vocal music has been integral to human expression since the earliest civilizations, serving as a medium for communication, cultural transmission, and emotional articulation. In recent decades, the a cappella genre has expanded significantly alongside advancements in digital audio technologies, which now provide unprecedented control over vocal production through effects, plugins, and detailed spectral manipulation. As ensembles pursue increasingly sophisticated blends, producers often encounter challenges in distinguishing individual vocal sections—most commonly including soprano, mezzo-soprano, alto, tenor, baritone, and bass—within densely layered textures. This difficulty can complicate decisions related to processing, balance, and stylistic refinement throughout an arrangement.

Contemporary audio tools rely predominantly on auditory perception, which may not always yield precise information regarding frequency overlap, sibilance distribution, or sectional balance. To address this limitation, the present thesis introduces a surround vocal spectrum analyzer designed specifically for multi-part vocal music in a surround sound mixing context. The plugin provides a visual representation of section-based spectral content, enabling engineers and arrangers to assess blend quality, identify problematic consonant clusters, and make more informed mixing decisions. By integrating visual analysis with traditional listening practices, this tool aims to support a more accurate and efficient workflow for vocal production.

Review of the State of the Art

An analysis of existing works and foundational research on stereo imaging, digital audio processing, and vocal ensemble blend.

The world of a cappella and choral music represents a unique creative niche, one that explores countless techniques and recording approaches to capture the richness of the human voice. Unlike the expansive textures produced by diverse instruments such as woodwinds, strings, brass, or percussion, vocal-based music relies entirely on the expressive potential of the human voice to create contrast and depth. Previous research has examined how singers adjust resonance and vocal tract alignment to achieve blend and tonal cohesion within ensembles, highlighting the acoustic interdependence between voices in shared space¹. The timbre of each voice is shaped by factors such as vocal placement, dynamic control, and inherent physical characteristics unique to each singer. In recording and mixing a cappella music, producers have a wide range of tools to shape and enhance the sound. Panning allows individual vocal parts to be positioned across the stereo field, helping listeners perceive each line as a distinct element within the mix. Additionally, clip gain and volume automation are widely used to adjust the balance of focus towards different parts in various sections of a song to provide clarity to more important parts and introduce texture to the music. The perceptual separation of vocal parts within a stereo field aligns with principles of spatial localization and interaural time difference described in Gerzon's² foundational work on stereo image stability. Percussive syllables—particularly sharp consonants like “t” or “k”—can be

¹ Woodruff, Neal Wayne. *The acoustic interaction of voices in ensemble: An inquiry into the phenomenon of voice matching and the perception of unaltered vocal process*. The University of Oklahoma, 2002.

² Gerzon, Michael. "Stabilising stereo images." *Studio Sound* 16, no. 12 (1974).

emphasized to create rhythmic clarity and guide the listener's focus between moving vocal lines. Similar perceptual dynamics have been observed in studies of spatial audio mixing in virtual environments, where transient localization significantly contributes to auditory scene clarity³.

Tuning and equalization techniques can also be applied to accentuate specific frequency ranges, allowing each voice type to shine: brighter EQ curves may highlight soprano and mezzo parts, while deeper, warmer tones can emphasize baritone and bass voices. Such spectral adjustments correspond to differences in frequency-domain energy distribution measurable through the Fast Fourier Transform, a mathematical process that converts time-domain audio signals into their constituent frequency components. FFT-based analysis has been widely applied in both music audio analysis and scientific signal processing.^{4,5}

However, in high-blend environments of mixing, it can sometimes be difficult for the producer to differentiate where in surround space an audio source is coming from and make informed decisions based on the ear's interpretation alone. While professional metering tools such as iZotope Insight⁶ and Nugen Visualizer⁷ assist with directional and

³ Bargum, Anders Riddershom, Oddur Ingi Kristjánsson, Péter Babó, Rasmus Eske Waage Nielsen, Simon Rostami Mosen, and Stefania Serafin. "Spatial audio mixing in virtual reality." *Sonic Interactions in Virtual Environments* 269 (2023): 269-302.

⁴ Alessio, Silvia Maria. "Digital signal processing and spectral analysis for scientists: concepts and applications." (2015).

⁵ Li, Yi. "Application of Spectrum Analysis Technology in Music Audio Analysis." *Advances in Multimedia* 2022, no. 1 (2022): 2196154.

⁶ iZotope. "Insight 2 - Intelligent Metering Plug-in | iZotope," 2026.
https://www.izotope.com/products/insight?srltid=AfmBOopsR9-O4pX2PVFRYH-twZsXmsL4eX7uEd6XM_PtHYIZ-k7IROik.

⁷ NUGEN Audio. "Surround Suite," 2018. <https://nugenaudio.com/surroundsuite/>.

spectral monitoring, they are designed primarily for instrumental or full-range mixes and offer limited functionality for vocal ensemble visualization.⁸ Furthermore, studies on ensemble blend suggest that psychoacoustic masking and resonance overlap can obscure spatial clarity in densely-layered voices.⁹

The Surround Vocal Spectrum Visualizer (SV2) aims to provide a visual aid for the producer to associate vocal panning not just with audio interpretation, but also with visual representation of the surround space relative to the listener in order to make more informed decisions on placement and balance of individual voice parts in a mix. Developed in C++ using the JUCE framework, an open-source C++ framework widely used for professional audio plugin development, and exported via VST3, AU, and AAX plugin formats, the plugin is compatible with major Digital Audio Workstations (DAWs) while following established practices for real-time digital signal processing (DSP) visualization and plugin architecture. This functionality is achieved through two primary components: a surround spatial diagram and a frequency spectrum display.

⁸ Ferdian, Robby, Irdhan Epria Darma Putra, Hengki Armez Hidayat, Ressay Fitria, and Revellame Revellame. "Analysis of Timbre in Melodic Arrangements of Popular Minang Songs Using a Spectrum Analyzer." *Journal of Urban Society's Arts* 10, no. 1 (2023): 32-42.

⁹ Ternström, Sten, Harald Jers, and John Nix. *Group and ensemble vocal music*. Oxford University Press, 2012.

Description

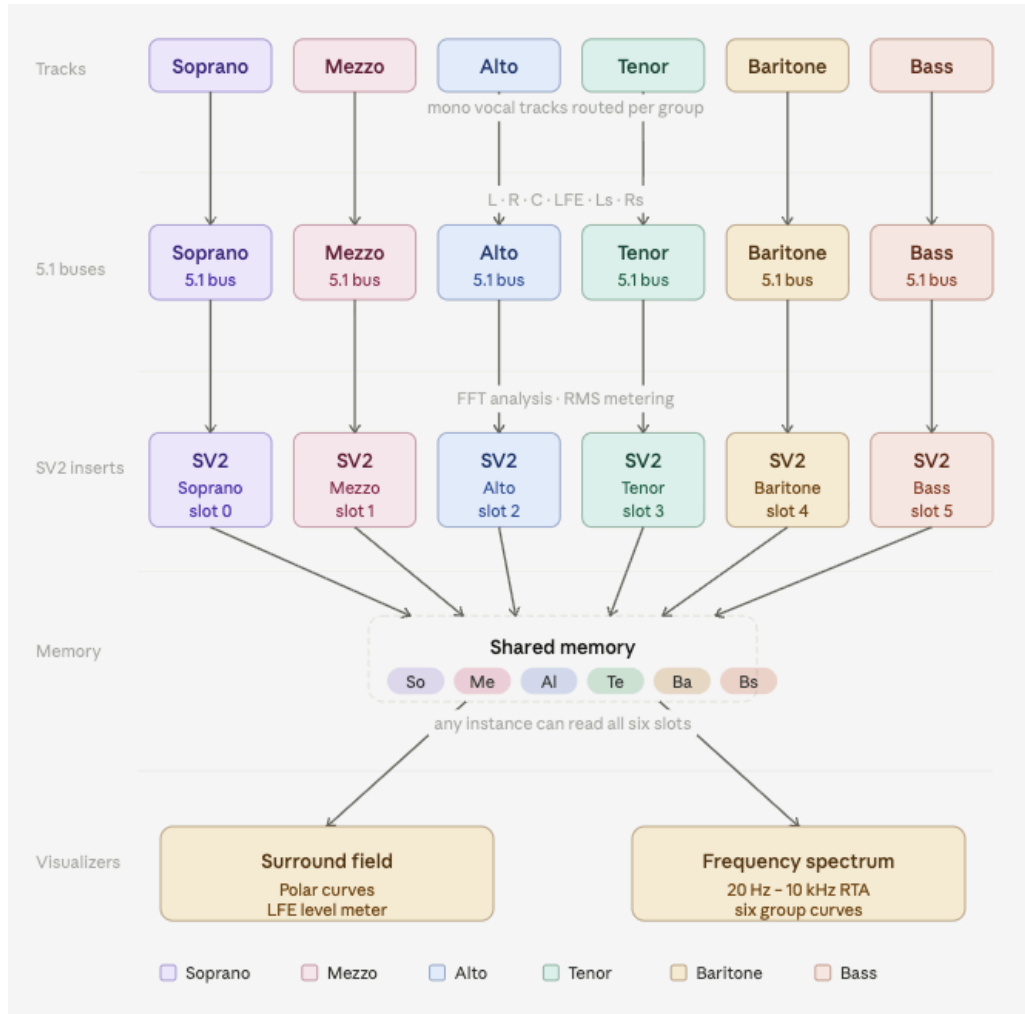


Figure 3.1. Signal flow illustrating routing from vocal subgroup buses in the DAW into separate SV2 plugin instances for real-time spatial and spectral visualization.

Using a surround field diagram, each vocal part—such as soprano, mezzo, alto, tenor, baritone, and bass—can be represented as a radius positioned relative to the listener at the center. The radius of each node indicates its relative gain, while the angular position corresponds to its placement in the surrounding panorama.

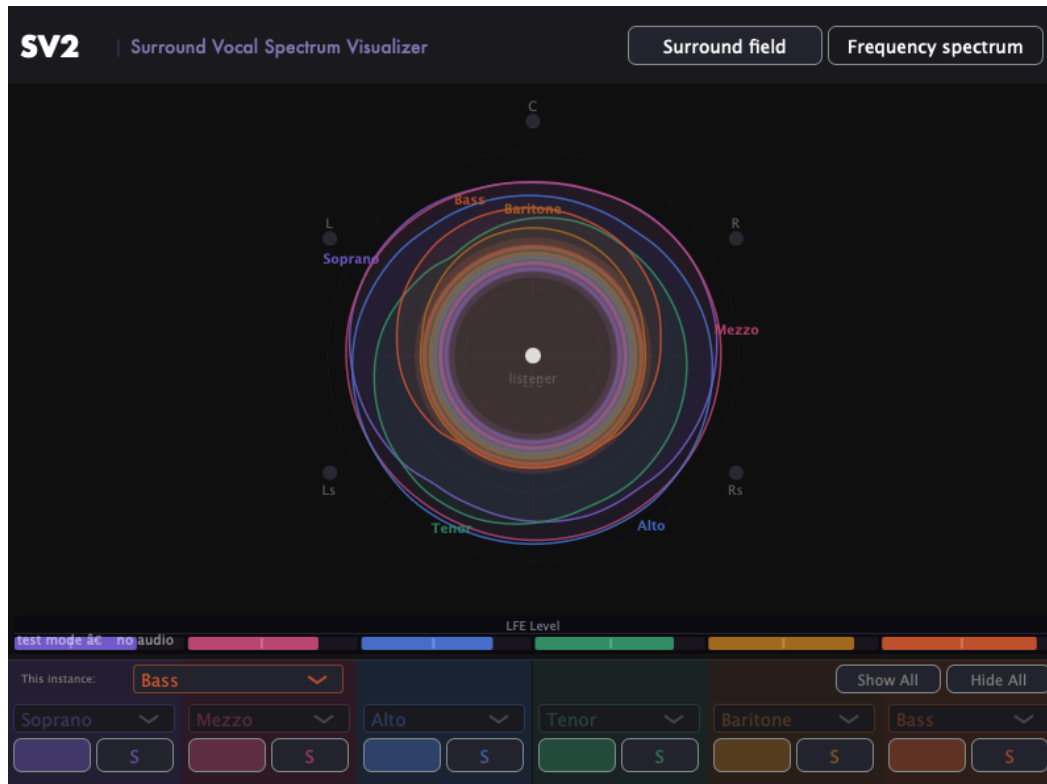


Figure 3.2. SV2 surround visualization window showing vocal subgroup placement within a 5.1 surround field. Node radius represents relative gain, while angular position corresponds to channel panning.

This visualization assists producers and engineers in making informed decisions about spatial balance, and allowing them to highlight different sections of a song and identify directional placement and relative balance of subgroup volume within the mix. This visualization approach parallels techniques in spatial audio research that enhance user perception of source localization and panning accuracy through dynamic visual mapping¹⁰. The surround window uses gain per channel observed from the subgroup to depict the radius width - monitoring Left, Center, Right, Leftside Rear, and Rightside Rear channels. A separate monitor on the lower portion of the plugin window displays

¹⁰ Gerzon, Michael. "Stabilising stereo images." *Studio Sound* 16, no. 12 (1974).

LFE channel sends, as this channel has no directional placement and is sent to the subwoofer component of the 5.1 audio format.¹¹ Because vocals are generally not sent to the LFE, the main channel that implements this meter is the Bass vocal subgroup to show the amount of low-end frequencies sent to the subwoofer outputs.

A separate second window within the plugin uses a frequency spectrum graph, analogous to an equalizer spectrum display, allowing engineers to observe the distribution of frequencies across all vocal parts.

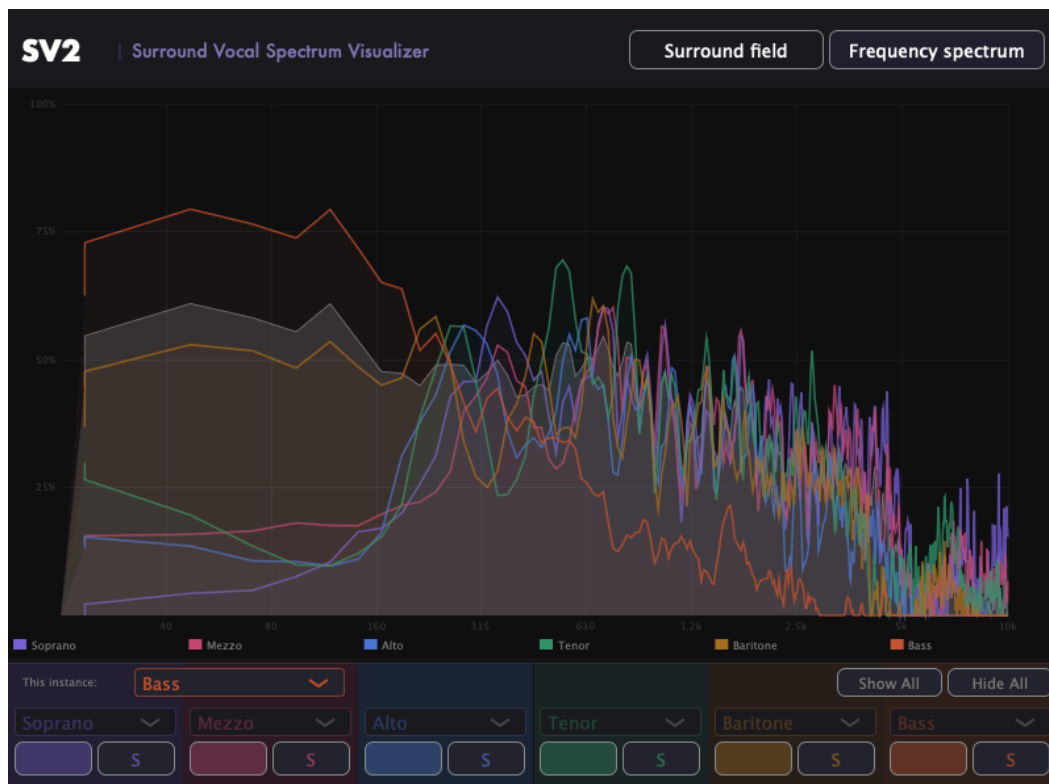


Figure 3.3. SV2 frequency spectrum visualization displaying FFT analysis of multiple vocal subgroups for monitoring spectral balance and masking.

¹¹ Rumsey, Frazncis. "Surround sound." *Immersive sound*, Routledge (2017): 180-220.

This visualization helps analyze tonal differences between groups of varying vocal ranges—such as the brighter timbres of upper voices versus the resonant tones of lower voices. Because each singer’s voice is shaped by unique biological and experiential factors like gender, physiology, vocal training, and age, the resulting timbral diversity contributes to the ensemble’s collective sound.¹² Spectral graphs of this nature, using FFT analysis, have been widely applied in both music signal analysis and ensemble acoustics studies to quantify brightness, resonance, and harmonic distribution. By referencing these frequency profiles, engineers can refine the overall mix to achieve a cohesive and intentional sound, whether seeking greater tonal consistency or highlighting the ensemble’s natural contrasts. Such an approach integrates artistic decision-making with perceptual principles related to spectral balance, clarity, and ensemble blend.

¹² Meerschman, Iris, Evelien D'haeseleer, Heleen Cammu, Imke Kissel, Tine Papeleu, Clara Leyns, Julie Daelman et al. "Voice quality of choir singers and the effect of a performance on the voice." *Journal of Voice* 39, no. 2 (2025): 561-e1.

Both visualization windows incorporate a set of control groups that enable selective monitoring of user-defined combinations of vocal subgroups.

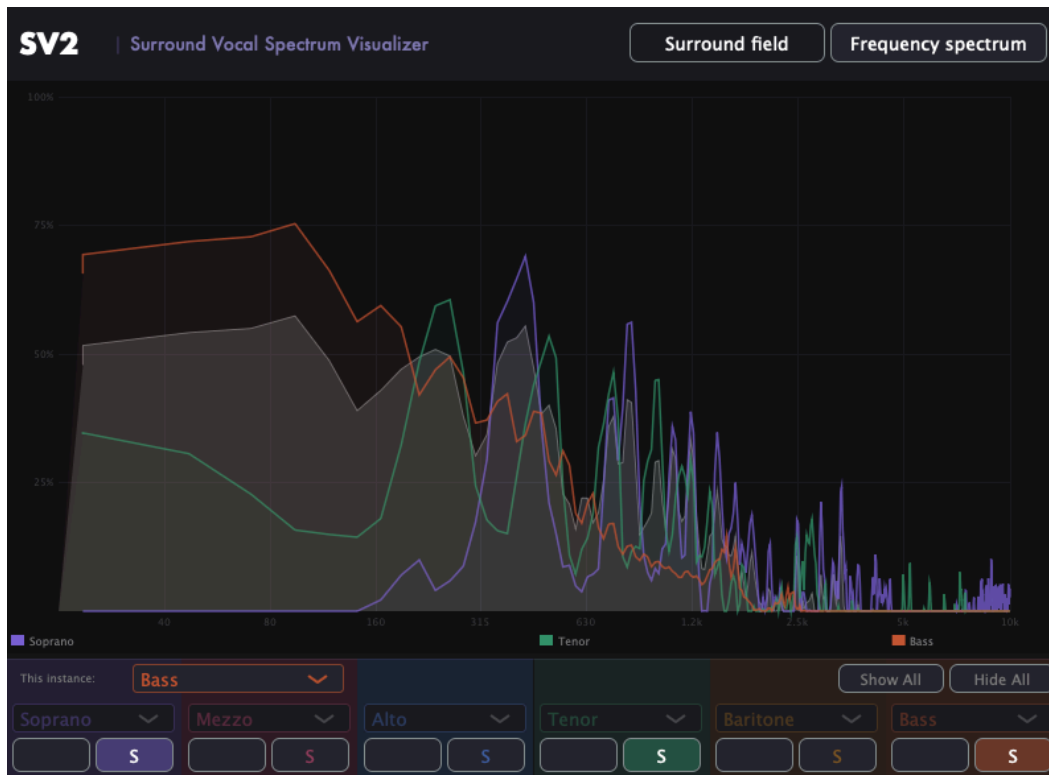


Figure 3.4. Shared control bar used for routing selection, subgroup visibility, solo functions, and global display controls.

Shared across both interfaces is a persistent control bar located at the bottom of the plugin window. This panel contains a routing selection menu that allows users to assign the plugin instance to a designated subgroup within the session. Additionally, individual visibility controls enable each subgroup to be displayed or hidden independently, while solo functions allow users to isolate a specific subgroup within the visual representation. Global “Show All” and “Hide All” controls provide efficient management of all subgroup displays simultaneously. These controls facilitate comparative analysis of selected vocal sections, allowing users to evaluate the spatial

distribution and spectral characteristics of related or contrasting voice groups within a mix. By enabling flexible combinations of subgroup visualizations, the interface supports more informed assessment of ensemble balance, blend, and overall mix cohesion.

Innovative Aspects

One of the primary innovative aspects of this project is its focus on supporting mixing decisions through visual feedback. Traditional mixing tools typically present information about individual aspects of a mix, such as signal level, frequency content, or stereo position. While these tools are effective for their intended purposes, engineers are often required to mentally combine information from multiple displays in order to understand how different vocal groups interact within a larger arrangement. SV2 was designed to reduce this disconnect by bringing spatial and spectral information together into a unified workflow. By providing an immediate visual representation of multiple vocal subgroups simultaneously, the software helps users identify potential balance issues and make more informed mixing decisions throughout the production process.

A second innovative aspect is the visual representation of vocal placement within surround space. In many mixing environments, spatial placement is communicated through panning controls, numerical values, or conventional metering systems. These approaches provide precise information but can make it difficult to quickly interpret the overall relationship between multiple vocal sections. SV2 instead presents vocal groups

as visual elements positioned relative to the listener within a surround field. The location of each subgroup reflects its placement within the mix, while its visual size corresponds to its relative level. This allows engineers to understand the spatial organization of an ensemble at a glance and identify situations where vocal groups may be competing for the same perceptual space.

The project is also innovative in its combination of spatial and spectral analysis within a single software tool. While frequency analyzers and spatial metering tools already exist independently such as iZotope Insight¹³ and Nugen Visualizer¹⁴, they are rarely integrated in a way that specifically supports ensemble vocal production. The surround visualization window allows users to observe the placement and balance of vocal groups, while the frequency spectrum window provides insight into the tonal characteristics and frequency distribution of those same groups. Together, these visualizations offer a more complete perspective on the mix, helping engineers evaluate not only where sounds are located, but also how they interact spectrally. This combined approach can assist in identifying masking, improving clarity, and achieving a more cohesive overall blend.

Another important aspect of the project is its emphasis on workflow efficiency and long-term project management. Music production projects often span weeks or months, making it difficult for engineers to immediately recall the reasoning behind previous mixing decisions when returning to a session. Rather than relying solely on

¹³ iZotope. “Insight 2 - Intelligent Metering Plug-in | iZotope,” 2026. https://www.izotope.com/products/insight?srsId=AfmBOopsR9-O4pX2PVFRYH-twZsXmsL4eX7uEd6XM_PtHYIZ-k7IROik.

¹⁴ NUGEN Audio. “Surround Suite,” 2018. <https://nugenaudio.com/surroundsuite/>.

listening and memory, SV2 provides a consistent visual reference for subgroup placement and balance. This allows users to regain context more quickly and compare current decisions against previously established relationships within the mix. Although the software does not replace critical listening, it serves as an additional reference point that can help maintain consistency throughout extended production cycles.

The project also explores applications beyond traditional stereo mixing workflows. While many visualization tools are designed primarily for stereo production, SV2 was developed with multichannel and surround audio environments in mind. The plugin's surround display is capable of representing subgroup relationships within a 5.1 listening environment, allowing engineers to better understand how vocal sections are distributed throughout the sound field. As immersive audio formats continue to become more common across music, film, gaming, and other media, tools that provide intuitive visual feedback for multichannel productions may become increasingly valuable. The design of SV2 reflects this shift by focusing not only on stereo placement, but also on the broader spatial relationships present in surround formats.

Finally, the project demonstrates the practical implementation of these concepts within a professional audio software environment. Rather than existing solely as a research concept or prototype, SV2 was developed as a fully functional audio plugin using C++ and the JUCE framework. The software was designed for deployment in industry-standard plugin formats including VST3, Audio Unit (AU), and AAX, enabling compatibility with major digital audio workstations such as Logic Pro and Pro Tools.¹⁵

¹⁵ Albouy, Adrien, and Stéphane Letz. “Faust Audio DSP Language for JUCE.” In *Proceedings of the Linux Audio Conference*, 61–68. Saint-Étienne, France, 2017.

This allowed the project to bridge the gap between academic exploration and practical application, resulting in a tool that can be incorporated directly into real-world production workflows.

For the author, this project also represented a significant departure from previous work in both scope and technical complexity. Earlier projects had focused primarily on music production, recording, and rapid prototyping within environments such as Max/MSP. The project also provided the author's first extensive experience working in Pro Tools with multichannel and immersive audio workflows, developing proficiency in surround mixing techniques, Dolby Atmos concepts, and 5.1 routing, automation, and monitoring practice that directly informed the plugin's spatial visualization design.

Developing SV2 required independently designing, implementing, testing, and deploying a professional audio software product from the ground up. The project integrated concepts from audio engineering, digital signal processing, software architecture, and user interface design into a single application. This transition from creative audio production to professional software development marked a substantial expansion of technical capability and demonstrated the ability to translate an original research concept into a practical tool for industry use. The combination of audio visualization, multichannel support, and cross-platform plugin deployment represents a meaningful contribution to both the technical and creative aspects of modern vocal music production.

New Skills Acquired

Throughout the development of this project, substantial growth occurred across several areas including surround sound production, technical research and project management, audio software development, and digital signal processing. The interdisciplinary nature of the project required the integration of concepts from music production, computer science, acoustics, and software engineering, resulting in the acquisition of both technical and creative skill sets.

One significant area of growth involved surround sound and immersive audio production. Through coursework in Advanced Topics in Audio Engineering and practical work within Dolby Atmos and multichannel Pro Tools environments, the project developed proficiency in surround mixing workflows beyond traditional stereo production. This included multichannel routing, subgroup organization, bus and send management, automation of spatial positioning and level changes, and the use of immersive audio monitoring techniques. Exposure to surround sound theory and spatial audio concepts strengthened understanding of how listeners perceive directional information and how spatial relationships can be utilized as a creative mixing tool.^{16 17} These experiences directly informed the design of the plugin's surround visualization window and its approach to representing subgroup placement within a multichannel environment.

¹⁶ Rumsey, Francis. "Surround sound." *Immersive sound*, Routledge (2017): 180-220.

¹⁷ Arteaga, Daniel. "Introduction to Ambisonics." Escola Superior Politècnica, Universitat Pompeu Fabra, Barcelona, Spain, 2015.

A second area of development involved technical research and project management. The project required translating a practical workflow challenge—understanding and monitoring relationships between vocal groups within a mix—into a set of technical requirements that could be implemented through software. This process involved reviewing literature related to vocal acoustics, timbre analysis, surround sound production, and audio visualization to identify gaps within existing workflows and establish a foundation for the project's design. Through this research process, the project strengthened skills in evaluating existing technologies, defining project scope, and balancing technical feasibility with artistic goals. The ability to move from an abstract concept to a functional prototype represented a significant advancement in both research methodology and long-term project planning.

The project also provided extensive experience in audio plugin development. Prior to this work, software development experience primarily involved general programming and prototyping environments such as Max/MSP. Developing SV2 required learning C++ and the JUCE framework, a widely adopted platform for professional audio software development.¹⁸ This process built proficiency in plugin architecture, graphical user interface design, real-time audio processing constraints, and deployment workflows. It also strengthened familiarity with version control practices using Git and GitHub, debugging within Visual Studio Code, and maintaining project organization throughout an extended software development cycle.

¹⁸ Albouy, Adrien, and Stéphane Letz. “Faust Audio DSP Language for JUCE.” In *Proceedings of the Linux Audio Conference*, 61–68. Saint-Étienne, France, 2017.

Particular attention was devoted to understanding industry-standard plugin formats and their role within professional production environments. The project was compiled and tested in VST3, Audio Unit (AU), and AAX formats to ensure compatibility across major digital audio workstations. While VST3 and AU deployment primarily involved framework configuration and testing, support for AAX development required a deeper understanding of the Pro Tools ecosystem, including developer licensing, plugin validation, and deployment procedures. Obtaining and utilizing an Avid developer license provided practical insight into professional software distribution workflows and cross-platform compatibility considerations.

The development process further strengthened knowledge of digital signal processing and audio analysis techniques. A significant portion of the project focused on understanding how audio information is represented digitally and how that information can be translated into meaningful visual feedback. This included studying frequency-domain analysis through the Fast Fourier Transform (FFT), a mathematical process that converts time-domain audio signals into their constituent frequency components. FFT analysis forms the foundation of many modern spectrum analyzers and audio visualization tools because it enables the measurement and display of frequency content in real time.¹⁹ ²⁰ Implementing and testing FFT-based visualizations deepened understanding of spectral balance, harmonic content, frequency masking, and the relationship between signal analysis and user interface design.

¹⁹ Alessio, Silvia Maria. *Digital Signal Processing and Spectral Analysis for Scientists: Concepts and Applications*. Cham: Springer, 2015.

²⁰ Li, Yi. "Application of Spectrum Analysis Technology in Music Audio Analysis." *Advances in Multimedia* 2022, no. 1 (2022): 2196154.

Finally, the project provided experience in multichannel visualization and real-time graphical rendering. Developing the spatial visualization system required translating audio metadata such as channel level and spatial position into visual representations that could be interpreted quickly by users. This process involved balancing technical accuracy with usability, ensuring that visual information remained meaningful without overwhelming the mixing workflow. The combination of audio engineering, software development, and visual design represented a unique interdisciplinary learning experience and contributed substantially to the overall growth achieved during the project.

Challenges

As with many interdisciplinary projects that combine music production, software engineering, and user interface design, the development of SV2 presented a number of challenges throughout the academic year. Some of these obstacles were anticipated during the initial planning stages of the project, while others emerged only after development was underway. Addressing these challenges contributed significantly to both the technical development of the software and the overall learning outcomes of the project.

Expected Challenges

Because the primary goal of SV2 was to provide meaningful visual support for surround mixing decisions, designing a user interface that was both informative and

intuitive became a central challenge from the outset of the project. The visualizations needed to communicate spatial and spectral information clearly while avoiding excessive complexity or visual distraction. Determining which information would be most valuable to engineers, how it should be represented graphically, and how multiple vocal groups could be displayed simultaneously without overwhelming the user remained an important consideration throughout development.

A second anticipated challenge involved acquiring proficiency with professional audio plugin development using C++ and the JUCE framework. Although previous experience in computer science provided a foundation in programming concepts, developing a real-time audio plugin required learning a specialized software architecture distinct from traditional application development. This included understanding audio processing pipelines, plugin lifecycles, graphical user interface development, build systems, and deployment workflows. In addition, version control practices were implemented through Git and GitHub to support project organization, revision tracking, and iterative development.

The project also required a deeper understanding of digital signal processing concepts used to transform audio information into meaningful visual representations. The frequency spectrum display relies on Fast Fourier Transform (FFT) analysis, which converts time-domain audio signals into their constituent frequency components. This process enables the visualization of spectral content and frequency distribution, providing insight into harmonic balance and potential masking between vocal groups. Developing this functionality required research into signal analysis techniques and the

efficient implementation of real-time FFT processing within the constraints of an audio plugin environment.

Finally, maintaining efficient performance across both audio and graphical systems was recognized as a significant challenge. Real-time audio processing requires low latency and reliable operation, while dynamic visualizations must update continuously without negatively affecting system performance. Balancing computational efficiency, visual responsiveness, and overall user experience was therefore a key objective throughout the development process.

Unanticipated Challenges

One of the most significant unanticipated challenges emerged during the transition from early prototyping to full software implementation. The initial frameworking and planning for the plugin were created within the Max/MSP programming environment, building upon concepts and techniques learned through coursework in Nonlinear Structures and Advanced Topics in Music Technology during the academic year at Berklee Valencia. This included early demonstrations of object-based nodes and the translation of audio gain into visual radius values using built-in modules and incoming audio signals.

However, as the capabilities and limitations of the Max/MSP environment became clearer, it became apparent that while the software is highly effective for audio signal processing and rapid prototyping, it offers more limited functionality for

advanced visualization and user interface development. Because the visual component became a central aspect of the plugin's intended functionality, this presented a significant obstacle. Shifting from rapid prototyping to a fully developed C++ plugin required a substantial redesign of both the implementation and overall development approach. The original intention was to develop the plugin within Max/MSP and eventually export it through RNBO. However, many of the workflows and structures developed within the Max environment proved either redundant or difficult to translate efficiently into a production-level plugin architecture. This ultimately led to a complete redesign of the project using C++ and the JUCE framework, requiring much of the software to be rebuilt from the ground up.

Another unexpected challenge involved balancing functionality with simplicity. Numerous features were proposed during the early planning stages of the minimum viable product (MVP), with many potential additions intended to provide deeper analytical insight into the mixing process. However, as development progressed, the focus gradually shifted toward creating a simpler and more intuitive user experience. Determining which information was genuinely useful to the engineer while avoiding visual clutter and unnecessary features became a difficult balance to maintain. Many design decisions ultimately required prioritizing clarity and workflow efficiency over feature quantity, ensuring that the interface remained informative without becoming overwhelming.

In the final stages of development, the focus shifted toward software deployment and format compatibility. VST3 plugins are widely accepted across major digital audio workstations such as Logic Pro and Ableton Live, making them one of the most common

deployment formats for audio software developers. However, the AAX format used by Pro Tools introduces additional licensing and development requirements that make deployment significantly less straightforward. Learning the requirements for building, testing, validating, and packaging VST3, AU, and AAX plugins proved more complex than originally anticipated and required additional research and outreach to obtain the necessary resources.

To facilitate testing within Pro Tools, a Pro Tools Developer license was acquired through Avid, allowing unverified AAX plugins to be loaded within a sandbox development environment. While this environment enabled plugin testing, it also introduced several workflow limitations, including restrictions on saving sessions. As a result, multichannel monitoring and mixing tasks often required the use of Studio A for hardware playback and validation, while separate testing was conducted within the Pro Tools Developer environment. Throughout this process, each iteration of the plugin required rebuilding, exporting, and validating the software before functionality could be confirmed. Maintaining consistent version control through GitHub became increasingly important, allowing stable versions of the project to be preserved while new features, bug fixes, and interface revisions were developed and evaluated.

Future Ramifications and Developments

While the current implementation successfully demonstrates the core functionality of real-time object-based visualization for immersive audio workflows, several opportunities remain for further development and refinement. Future work will focus on expanding both the technical capabilities and practical usability of the plugin while maintaining the simplicity and workflow efficiency established during the MVP development process.

One area of future development involves expanding user interface customization options. Although the current interface provides a functional visualization environment, additional controls such as customizable group names, color palettes, visual themes, and layout preferences could allow users to tailor the interface to individual workflows and studio environments. Providing greater flexibility in how information is displayed may improve usability across a wider range of mixing and production scenarios.

Further performance optimization will also be an important priority. As immersive audio productions continue to increase in complexity, supporting larger numbers of simultaneous audio objects, plugin instances, and multichannel sources will become increasingly important. Future development will focus on improving processing efficiency, reducing visualization jitter, and enhancing overall system stability to ensure consistent real-time performance in demanding production environments.

Additional user testing represents another significant area for future work. While the current design has been guided by practical mixing workflows and iterative development, broader testing with engineers, producers, and students working in

immersive audio environments would provide valuable insight into how the plugin functions in a variety of professional contexts. Feedback gathered through these evaluations could inform interface refinements, feature prioritization, and workflow improvements that better address the needs of end users.

Looking beyond the immediate scope of the project, the underlying visualization framework could potentially be extended to support more advanced analytical tools for immersive audio production. Future iterations may explore additional spatial metrics, object relationship analysis, or enhanced monitoring capabilities that provide deeper insight into object placement and movement within three-dimensional audio environments. Such developments could further support engineers working in increasingly complex immersive mixing workflows while maintaining the project's primary goal of presenting information in a clear and intuitive manner.

Finally, future work will also consider the broader distribution and adoption of the plugin beyond the scope of this thesis. As SV2 has been developed and successfully deployed in VST3, Audio Unit (AU), and AAX formats, the software is capable of integration into existing professional production workflows. Continued development may include public release for evaluation within the audio engineering community, distribution through academic and educational institutions, or eventual commercial publication following additional testing and refinement. Sharing the plugin with producers, educators, and researchers would provide valuable real-world feedback while encouraging further discussion surrounding visualization techniques for immersive vocal production. Such dissemination would represent a natural progression from academic research to practical industry application.

Conclusions

This project explored the development of a visual analysis tool designed to support surround sound vocal mixing through the integration of spatial and spectral visualization. Existing mixing workflows rely heavily on auditory perception and a collection of separate metering tools, requiring engineers to interpret multiple sources of information when evaluating the balance and placement of vocal ensembles. The goal of SV2 was to investigate whether these relationships could be represented more intuitively through a unified visual interface that complements traditional listening practices.

Through the development of a fully functional audio plugin using C++ and the JUCE framework, the project successfully demonstrated the implementation of two primary visualization systems: a surround-field display for monitoring subgroup placement and balance, and a frequency spectrum analyzer for observing tonal relationships between vocal sections. Together, these tools provide engineers with an additional reference point for understanding how vocal groups interact within a multichannel environment. While the software is not intended to replace critical listening, it offers a visual framework that can support more informed decision-making throughout the mixing process.

The research conducted throughout the project highlighted the growing importance of immersive audio production and the increasing need for tools that help engineers navigate complex spatial environments. Although many existing plugins provide detailed information regarding level, frequency content, or spatial position independently, relatively few tools combine these perspectives in a manner specifically

tailored to ensemble vocal production. By focusing on the unique challenges of vocal blend, timbral variation, and spatial organization, SV2 contributes a specialized approach to audio visualization that addresses an underexplored area of music production technology.

Beyond the software itself, the project demonstrates the value of interdisciplinary problem solving at the intersection of music production, acoustics, digital signal processing, and software engineering. The process required translating a practical workflow challenge into a functional software solution, while balancing technical accuracy, usability, and artistic flexibility. The resulting plugin represents both a technical achievement and a practical exploration of how visual feedback can enhance creative decision-making within professional audio workflows.

Ultimately, SV2 serves as a proof of concept for the role that integrated visualization systems can play in modern music production. As immersive and multichannel audio formats continue to expand across music, film, gaming, and interactive media, tools that provide intuitive representations of complex spatial and spectral relationships may become increasingly valuable. The project therefore contributes not only a functional software tool, but also a foundation for future exploration into the relationship between audio perception, visualization, and creative workflow design.

Appendix I A: Gantt Chart

TASK NAME	DECEMBER					JANUARY				FEBURARY				MARCH					APRIL				MAY
	1	2	3	4	5	1	2	3	4	1	2	3	4	1	2	3	4	5	1	2	3	4	1
Initial Sourcing and Research																							
Development Frameworking																							
Frequency Analysis																							
Functional Surround Space Parser																							
Separate Patch Integration																							
GUI Modification																							
AAX Plugin Publication																							
Prototype 1																							
Prototype 2																							
Prototype 3																							

Appendix I B: Updated Budget

Item	Quantity	Proposed (Berklee)	Market Cost
HARDWARE/EQUIPMENT			
2024 Macbook Pro 14in M4	1	1650	1650
1TB External SSD	2	100	100
Audio Technica ATX-M40x		109	109
Neuman M-149	2	0	5000
Neumann U87	2	0	2900
Neumann TLM67	2	0	2200
AKG C414	2	0	900
Shure SM7B	2	0	550
UAD Apollo Twin X	1	0	1169
SOFTWARE & LICENSING			
Pro Tools Developer License	1	0	n/a
Pro Tools Studio	1	0	299
Logic Pro X	1	0	200
JUCE Framework	1	0	0
iZotope RX	1	0	399
MAX/MSP Student License	1	0	199
RNBO Export License	1	299	299
Ableton Live Suite	1	374	749
STUDIO			
Surround Sound Studio	tba	tba	tba
Project Studio Rooms	tba	tba	tba
RATES			
Ensemble Vocalists	18	0	30/hr
Recording Engineer	1	0	45/hr
Personal Production	1	0	40/hr
TOTAL COST:	n/a	\$0	

Appendix II: Generative AI

Following this link leads to the description of the use of generative AI in this paper.

https://drive.google.com/file/d/1n_BsAXoP4CWdNT72euK_f_zuAqoGJU5z/view?usp=sharing

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